



BKBPS TEACHERS

# Capstone Project Presentation



## Group Members:

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Presentation Tool: Canva

# Problem Statement



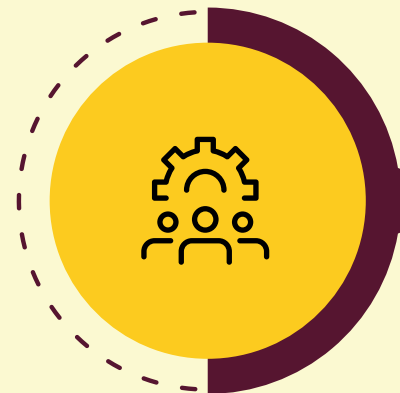
Many people use single-use plastic daily because it is convenient and cheap. How might we design affordable and sustainable alternatives or systems that encourage people to reduce their dependence on single-use plastic?



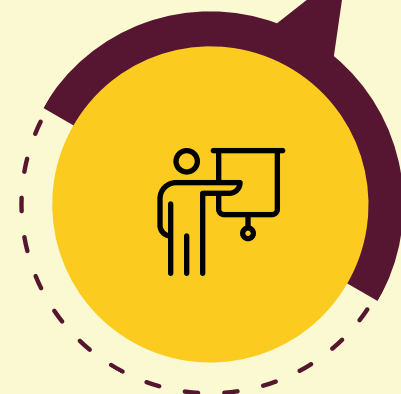
1. Many Interconnected Causes



2. Multiple Stakeholders



3. No Single "Correct" Solution

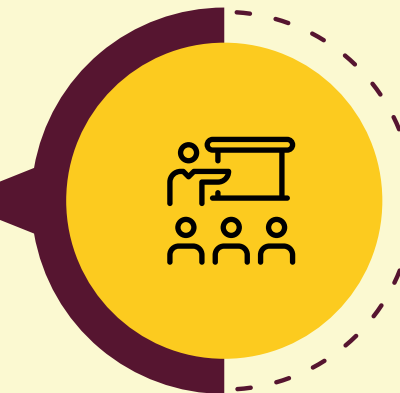


WHY IS IT A  
COMPLEX  
PROBLEM?

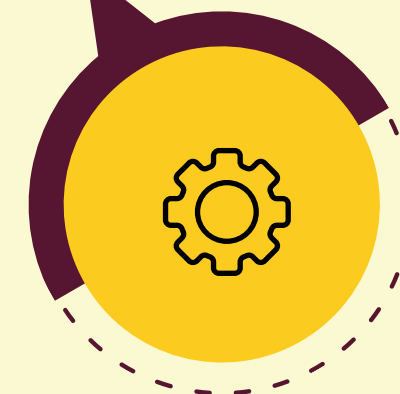
4. Behaviour Change is required



5. Trade-offs Between  
Cost, Convenience, and  
Sustainability



6. Solutions Must Work in  
Real Contexts





# By the end of the project, students will:

## **PRESENT**

a tested solution to reduce plastic waste in their school or community.

## **DEMONSTRATE**

the Design Thinking process from empathy to testing.

## **REFLECT**

on how individual actions can contribute to environmental sustainability.



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# Goal



Students will investigate the problem of plastic waste in their community and design practical, user-centered solutions that reduce the use and impact of single-use plastics.





**Table 1 – Project Planning (Design Thinking Model)**

| Phase     | Stage                       | Activities                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|-----------|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Scope     | Problem Exploration         | <ul style="list-style-type: none"><li>• Brainstorm sources of single-use plastic in daily life (bags, bottles, packaging, straws).</li><li>• Conduct waste audit in school and nearby market to identify major plastic waste sources.</li><li>• Study statistics on plastic waste generation in India.</li><li>• Create a <b>visual map of plastic usage areas</b> (home, school, shops, restaurants).</li><li>• Review case studies from the cleanest cities such as Singapore, Copenhagen and Tokyo.</li></ul> |
| Empathize | Stakeholder & User Research | <ul style="list-style-type: none"><li>• Conduct surveys among households, students and shopkeepers to understand plastic usage behaviour.</li><li>• Interview waste collectors to understand challenges in plastic recycling.</li><li>• Observe consumer habits in local markets and supermarkets.</li><li>• Create <b>stakeholder mapping</b> (consumers, businesses, government, recyclers).</li><li>• Develop an <b>empathy map</b> to understand motivations, needs and problems of users.</li></ul>         |
| Define    | Problem Analysis            | <ul style="list-style-type: none"><li>• Analyze survey data to identify the main causes of plastic dependency.</li><li>• Create <b>cause-effect (fishbone) diagram</b> to identify root problems.</li><li>• Identify gaps in current waste management systems.</li><li>• Compare waste practices in India with those of clean cities.</li><li>• Write a clear <b>problem statement</b> describing the challenge.</li></ul>                                                                                       |







# Project planning (DT Model)

| PHASE      | STAGE               | ACTIVITIES                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|------------|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Ideate     | Solution Generation | <ul style="list-style-type: none"><li>• Brainstorm innovative ideas to reduce plastic use.</li><li>• Study global solutions like bottle deposit systems, refill stations and waste-to-energy plants.</li><li>• Generate alternative materials (cloth bags, biodegradable packaging, reusable containers).</li><li>• Conduct group discussions to evaluate feasibility of solutions.</li><li>• Prepare concept sketches and system diagrams for selected ideas.</li></ul> |
| Prototype  | Model Development   | <ul style="list-style-type: none"><li>• Create a prototype such as <b>cloth bag bank, bottle return station, plastic collection box or refill station model</b>.</li><li>• Design posters, awareness materials and process flow diagrams.</li><li>• Develop a <b>working model demonstrating the plastic reduction system</b>.</li><li>• Test usability of the prototype with students and teachers.</li></ul>                                                           |
| Validation | Testing & Feedback  | <ul style="list-style-type: none"><li>• Implement the prototype in a small area (school or classroom).</li><li>• Collect feedback from users regarding convenience and effectiveness.</li><li>• Measure reduction in plastic waste during the trial period.</li><li>• Identify limitations and areas for improvement.</li></ul>                                                                                                                                          |







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# Project planning (DT Model)

PHASE

STAGE

ACTIVITIES

**Implementation**

Real  
Application

- Implement the improved system in school or nearby community.
- Install segregation bins and plastic collection stations.
- Introduce reusable bag systems and plastic return incentives.
- Conduct awareness campaigns and workshops.

**Report**

Documentation  
& Reflection

- Document research findings, survey results and global case studies.
- Analyze the effectiveness of implemented solutions.
- Present results through charts, graphs and photographs.
- Provide recommendations for scaling the solution to larger communities.





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# Scamper



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# Requirements & Specifications



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# Solution Model



# PHASE WISE PLANNING



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Your Attention

